

Deep Learning for Time Series

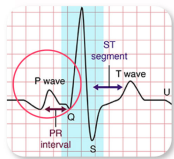
Session 5: Continuous-Time Models

Romain Tavenard

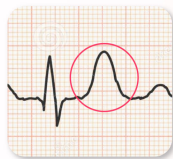
Motivation

Understanding ECG Waves

Types of **Abnormal** ECG



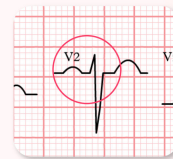
Arrhythmia



Ischemia/Infarction



Conduction Abnormalities

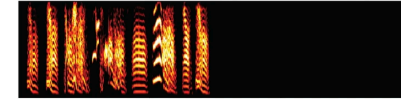


Hypertrophy/Enlargement

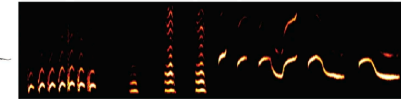
Source: sunfox.in blog



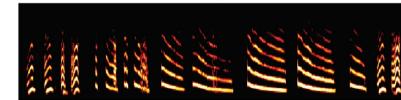
Zebra finch (*Taeniopygia castanotis*)



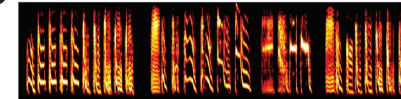
Long-tailed finch (*Poephila acuticauda*)



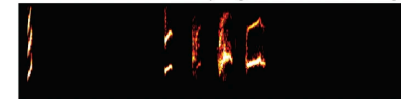
Double-barred finch (*Stizoptera bichenovii*)



Bengalese finch (*Lonchura striata* var. *domestica*)



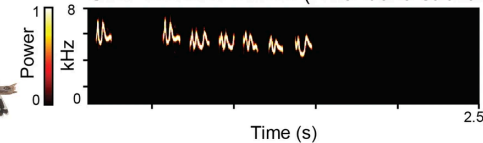
Red-billed firefinch (*Lagonosticta senegal*)



Red-cheeked cordon-bleu (*Uraeginthus bengalus*)



Gold-breasted waxbill (*Amandava subflava*)



Source: “Machine learning and statistical classification of birdsong link vocal acoustic features with phylogeny”

Motivation

- Time series are (almost always) discretization of continuous-time processes
- In real life, sensors fail
 - Missing data
 - Irregular sampling
- Basic neural architectures not tailored for such settings
 - Conv vs Recurrent vs Attention-based
 - Missing data imputation can help

⇒ **Can we build neural architectures that operate in continuous time?**

Neural ODEs

Assume the evolution of the forecast variable through time follows a system of the form:

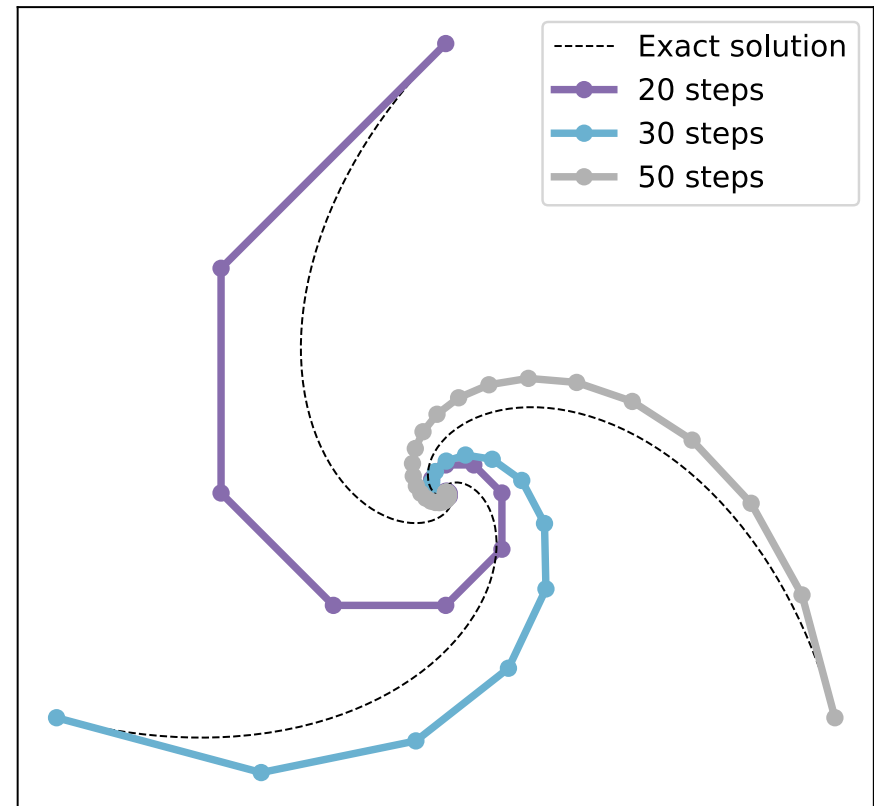
$$\begin{cases} \dot{x}(t) = f(x(t), t) \\ x(0) = x_0 \end{cases}$$

- Given f and x_0 , one can use an ODE solver to compute $x(t)$ for any t by approximating:

$$x(t) = x_0 + \int_0^t f(x(t), t) dt$$

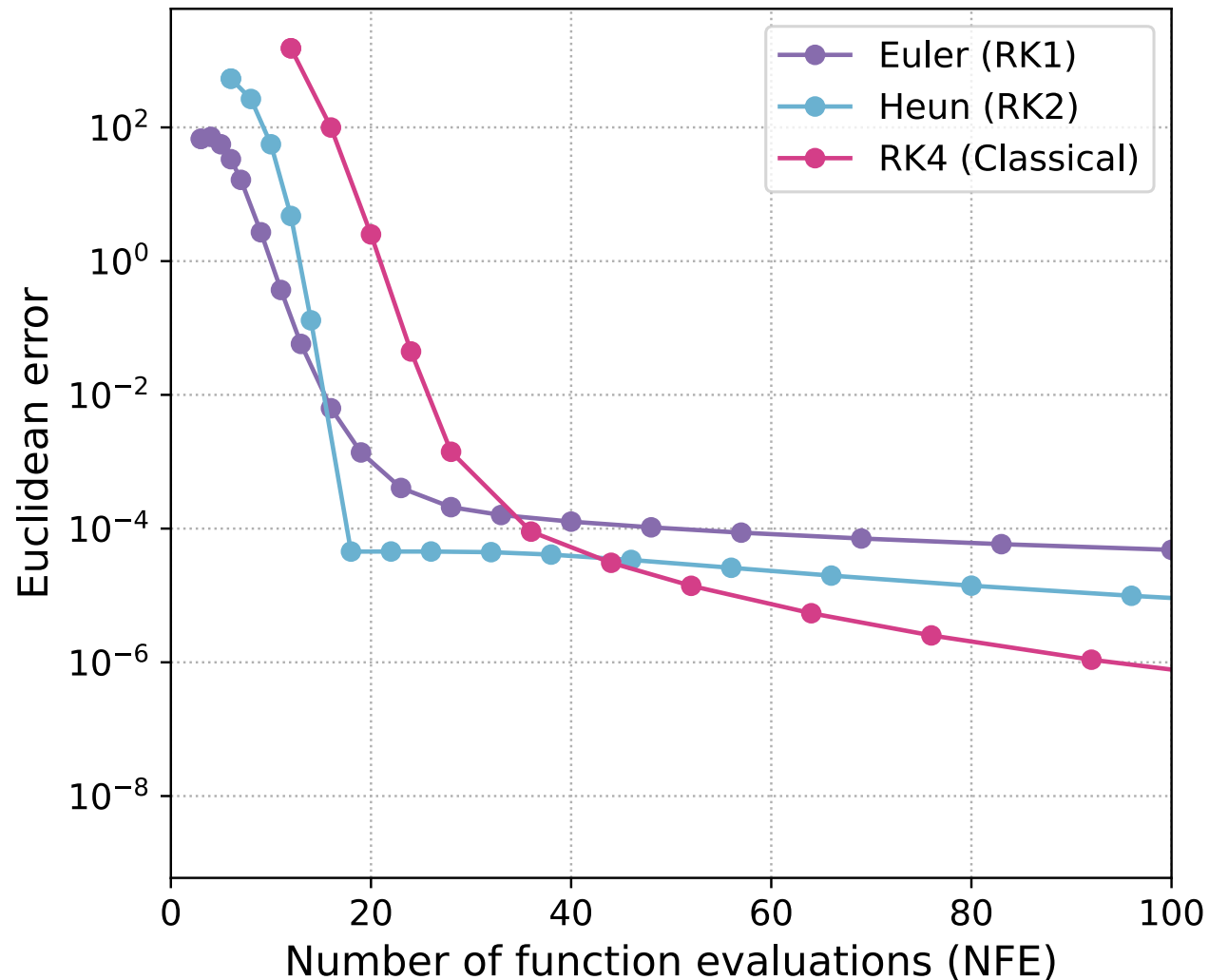
- Approximate the sequence of values:
 $[x(t + h), x(t + 2h), \dots, x(t)]$
- Using a Taylor expansion to compute $x(\tau + h)$ for $x(\tau)$
- First-order Taylor expansion gives the Euler method:

$$x(\tau + h) \approx x(\tau) + hf(x(\tau), \tau)$$

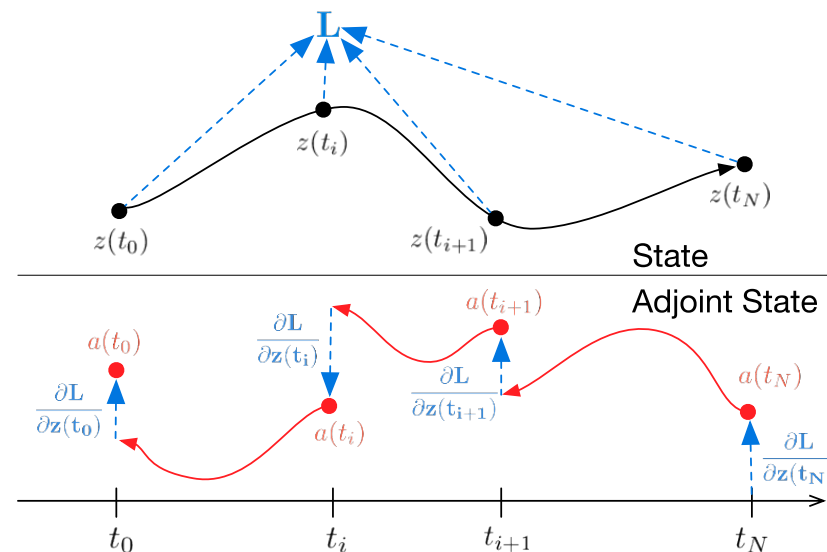


Solving a spiral ODE with a Euler scheme

- More advanced ODE solvers exist
 - Runge-Kutta methods (see below), adaptive step-size, ...



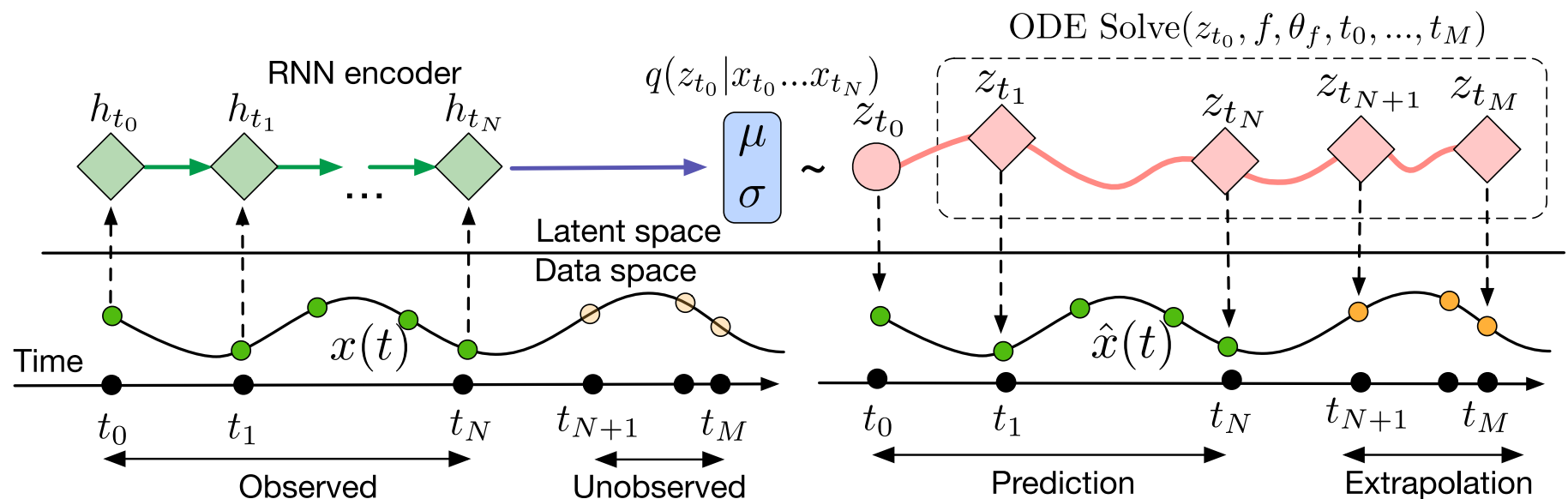
- ODEs for forecasting
 - Need to find the right f for our data
 - Let f be a neural network f_θ and train it on our forecasting task
 - Need to compute gradients through the solver



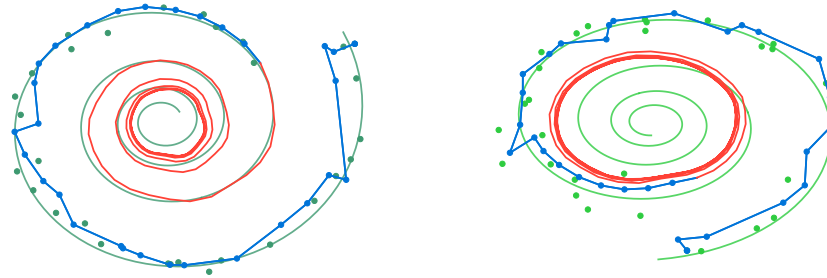
Source: “Neural ODEs”, NeurIPS’19

- Euler scheme: akin to a ResNet with shared weights

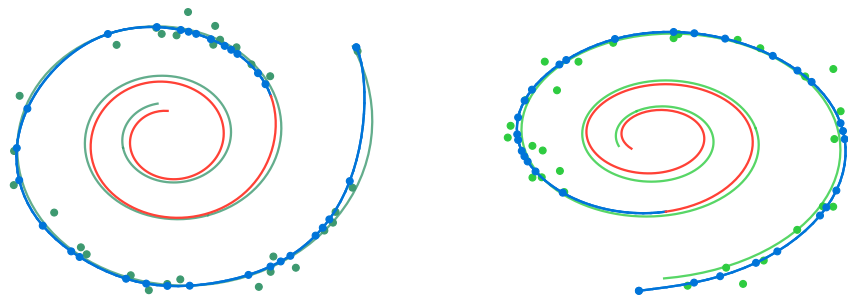
- Neural ODEs cannot operate in input space
 - Reconstruction from a single observation x_0 is a strong limitation
- Latent NODEs
 1. project in latent space to get higher-level information
 2. use RNN encoder to summarize information from the past



Source: "Neural ODEs", NeurIPS'19

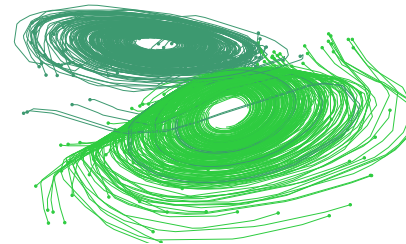


(a) Recurrent Neural Network



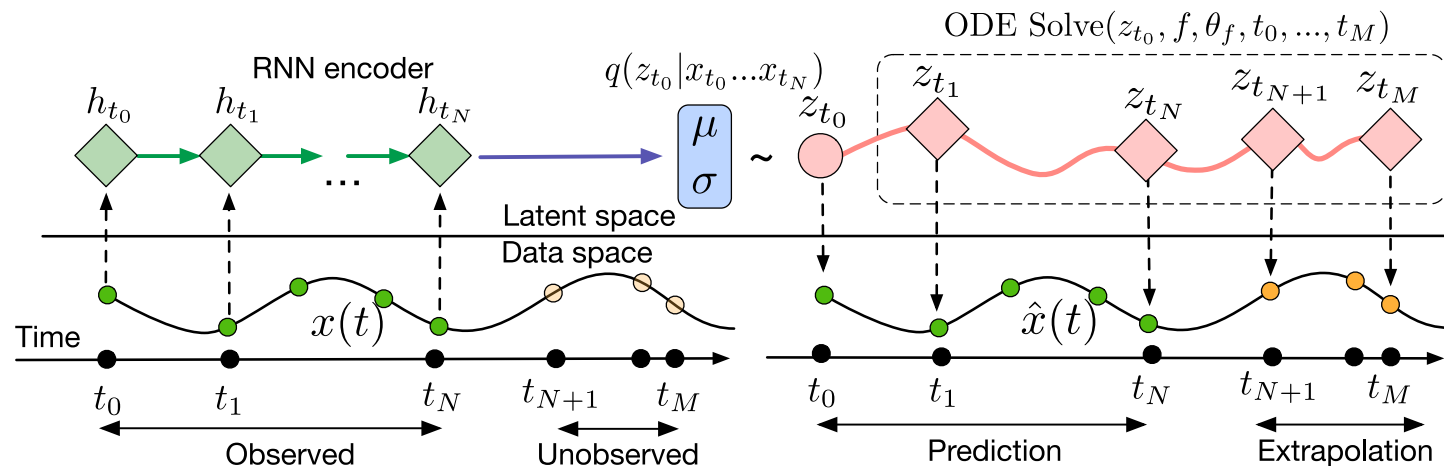
(b) Latent Neural Ordinary Differential Equation

— Ground Truth
● Observation
— Prediction
— Extrapolation

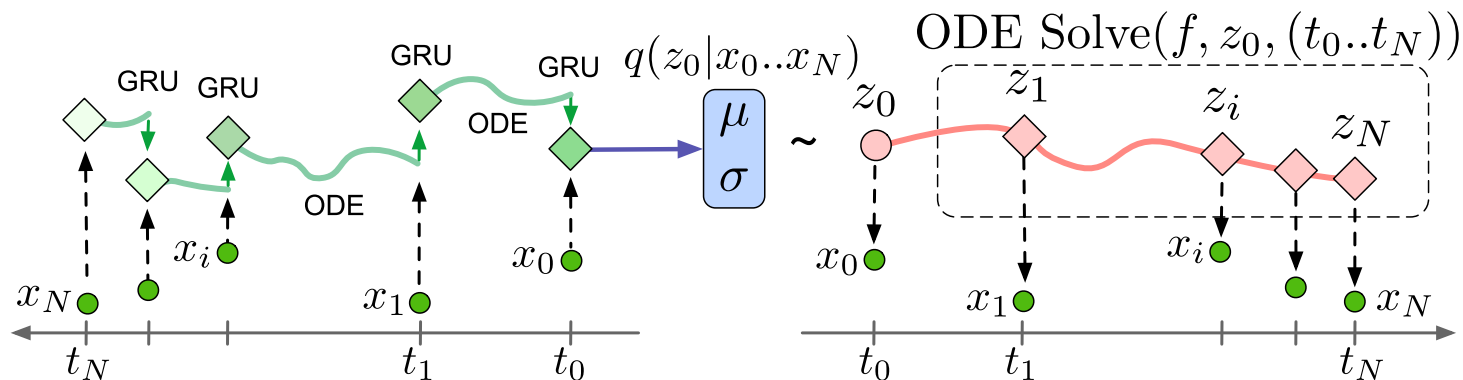


(c) Latent Trajectories

Source: “Neural ODEs”, NeurIPS’19



- RNN encoder above still assumes regular sampling



Source: “Latent ODEs for Irregularly-Sampled Time Series”, NeurIPS’20

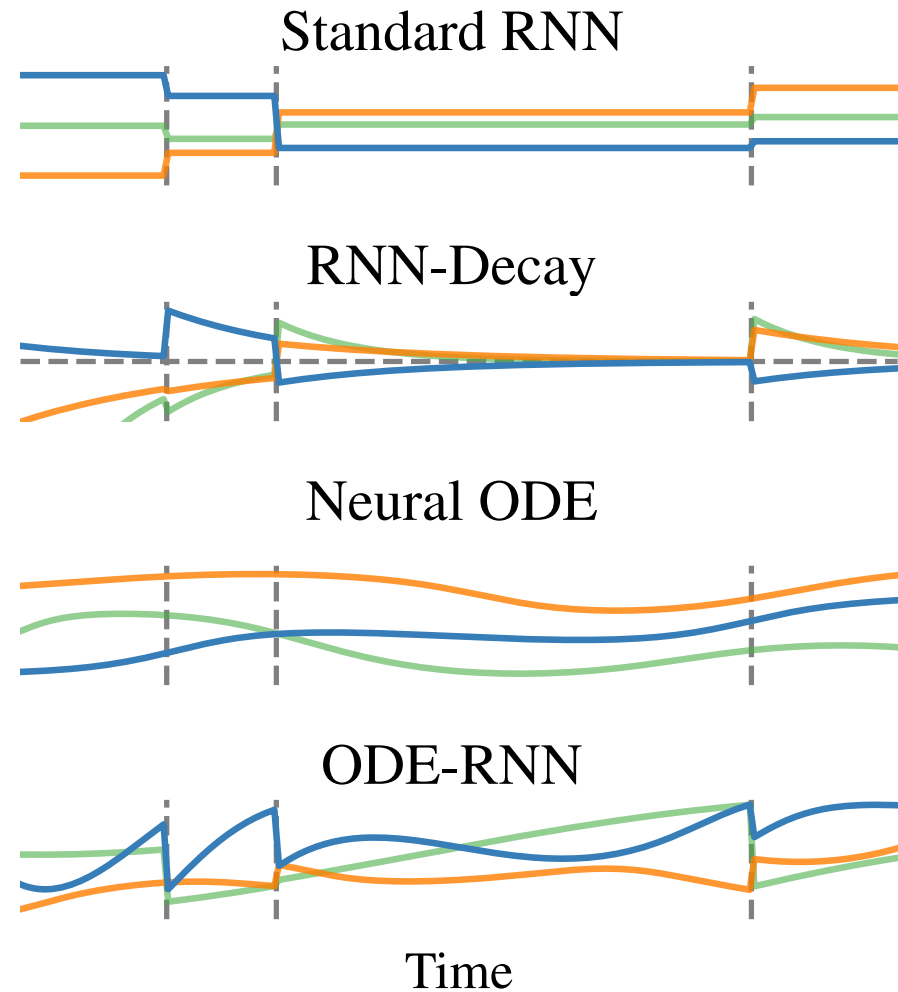
- Standard RNN cell update:

$$h_t = \text{RNN}(h_{t-\Delta t}, x_t)$$

- NODE-RNN cell update:

$$\tilde{h}_t = \text{ODESolve}(h_{t-\Delta t}, \Delta t)$$

$$h_t = \text{RNN}(\tilde{h}_t, x_t)$$



Source: “Latent ODEs for Irregularly-Sampled Time Series”, NeurIPS’20

Implicit Neural Representations

Implicit Neural Representations (INRs) model a signal as a continuous function of time:

$$\begin{aligned} f_{\theta} : \mathbb{R} &\rightarrow \mathbb{R}^d \\ t &\mapsto f_{\theta}(t) \end{aligned}$$

A common choice for f_{θ} is an MLP.

- Representing time as a 1D feature is a weak representation
- In practice, an INR's first layer is often a positional encoding
 - Fourier features (random or learnt):

$$h_1(t) = [\sin(\omega_1 t), \cos(\omega_1 t), \dots, \sin(\omega_K t), \cos(\omega_K t)]$$

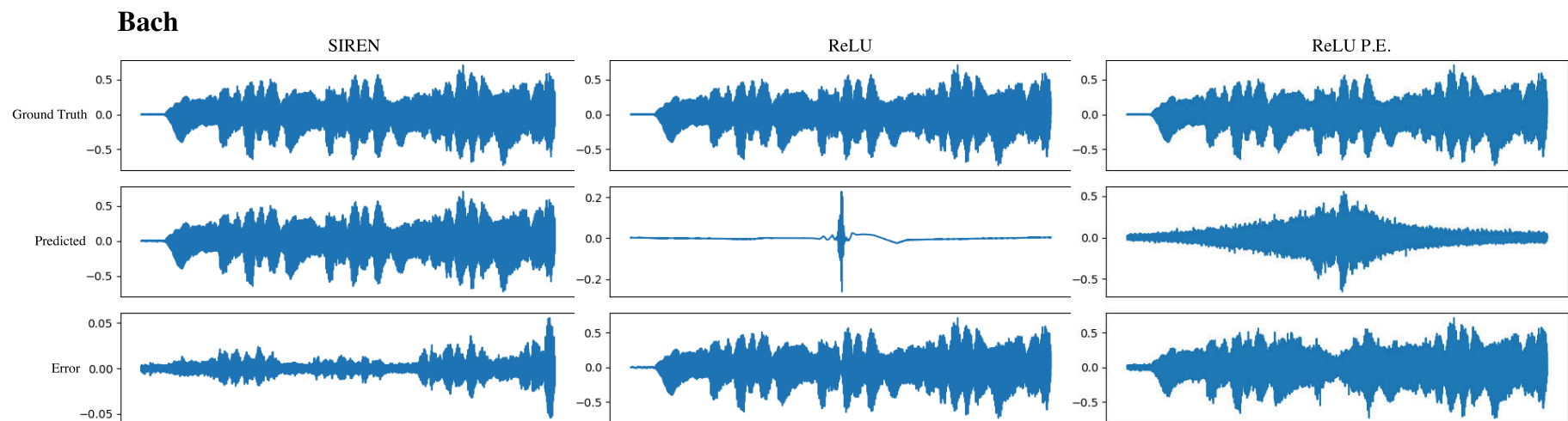
- Sine activated layer (W is a learnable vector):

$$h_1(t) = \sin(\omega_0 W t)$$

- Use sine activation functions:

$$\Phi(x) = \sin(W \cdot x + b)$$

- Acts as a learnable Fourier basis decomposition



Source: “Implicit Neural Representations with Periodic Activation Functions”,
NeurIPS’20

- Such INRs can learn to extrapolate a single time series:

$$f_{\theta} : \mathbb{R} \rightarrow \mathbb{R}^d$$
$$t \mapsto f_{\theta}(t)$$

- In practice, we have a dataset of time series, hence the formulation:

$$f_{\theta} : \mathbb{R} \times \mathbb{R}^p \rightarrow \mathbb{R}^d$$
$$t, z \mapsto f_{\theta}(t, z)$$

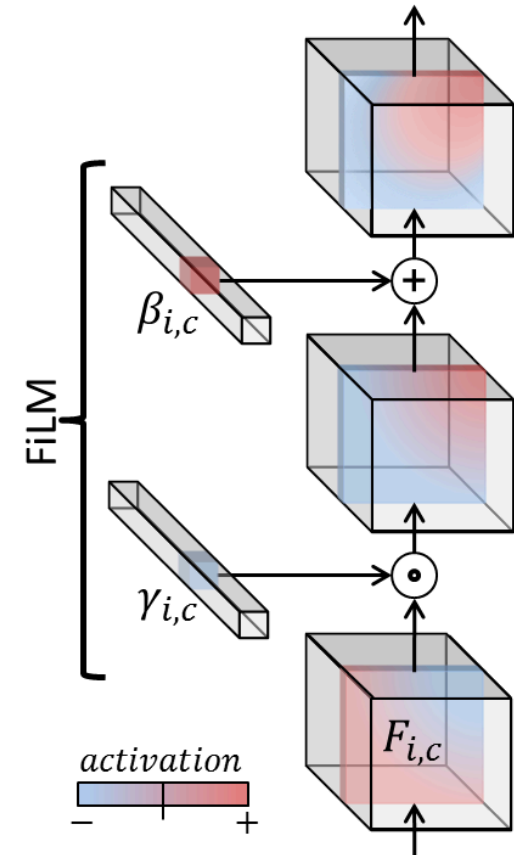
where z is a code summarizing the content of the time series

- **Modulated INRs**: z modulates the behaviour of f_{θ}
(activations or weights)

Feature-Wise Linear Modulation (FiLM)

1. The time series is encoded as z
2. A modulation network (shallow MLP) outputs modulation parameters $\gamma(z)$ and $\beta(z)$ for each INR layer
3. These parameters are used to modulate the INR **activations**

$$h^{\text{modulated}} = \gamma(z) \odot h^{\text{INR}} + \beta(z)$$



Source: “FiLM [...]”,
AAAI’18

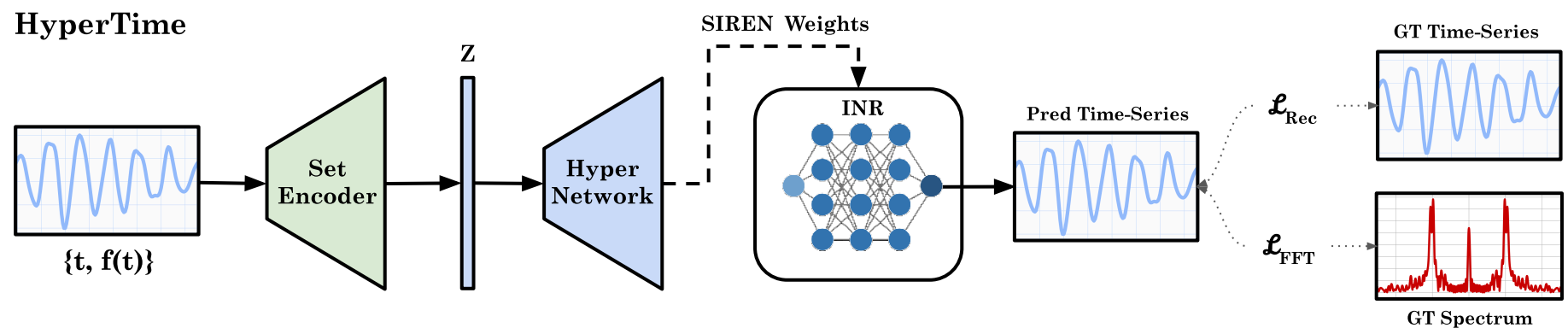
- Modulation at the INR **parameter** level:
 1. A hypernetwork learns per-parameter modulations $\psi(z)$
 2. The INR (hypernetwork) is now:

$$f_{\theta, \psi} : \mathbb{R} \times \mathbb{R}^p \rightarrow \mathbb{R}^d$$

$$t, z \mapsto f_{\theta + \psi(z)}(t)$$

HyperTime

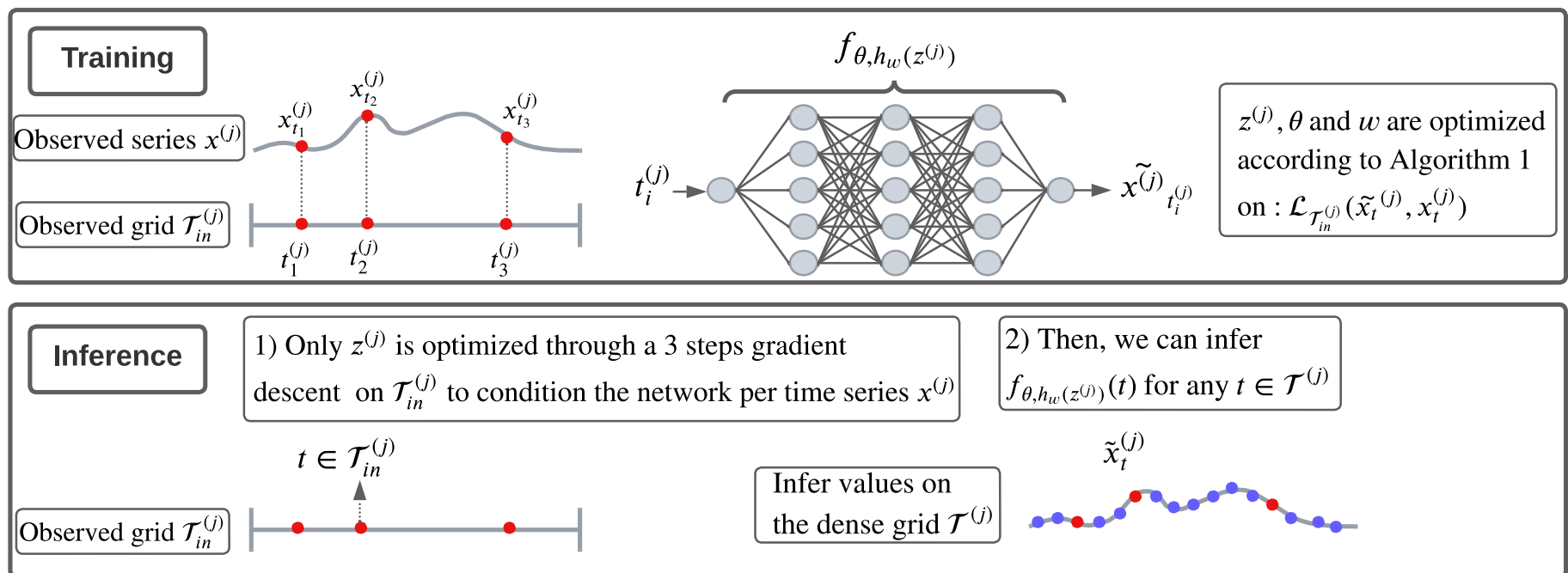
- Generates an encoding z per timestamp
- Requires global pooling along the time axis (or fixed number of observations per series)



Source: “HyperTime: Implicit Neural Representation for Time Series”, NeurIPS Workshop, 2022

TimeFlow

- Code z is optimized through few-step gradient descent (not output by an encoder)
 - no constraint on the input time grid, no need for pooling



Source: "Time Series Continuous Modeling for Imputation and Forecasting with Implicit Neural Representations", TMLR'24